

College Management Lightning Application – Complete Implementation & Troubleshooting Guide

Project Title

Develop a College Management Lightning Application to store and manage Student and Faculty records with validation rules.

Objective

Create:

- Student Object
 - Faculty Object
 - Validation Rules
 - Lightning App
 - Tabs
 - Data Validation System
-

Step 1: Create Student Object

Go to: Setup → Object Manager → Create → Custom Object

Details

Field	Value
Label	Student
Plural Label	Students
Record Name	Student Name

Save.

Step 2: Create Student Fields

Go to: Student → Fields & Relationships → New

Create these fields:

Field Name	Data Type
Roll Number	Number
Email	Email
Department	Picklist
Marks	Number
Joining Date	Date

Step 3: Department Picklist Values

Add values:

- Computer
- IT
- Mechanical
- Civil
- ENTC
- Electrical

Save.

Step 4: Create Faculty Object

Go to: Setup → Object Manager → Create → Custom Object

Details

Field	Value
Label	Faculty
Plural Label	Faculties
Record Name	Faculty Name

Save.

Step 5: Create Faculty Fields

Go to: Faculty → Fields & Relationships → New

Create:

Field Name	Data Type
Faculty ID	Number
Salary	Currency
Department	Picklist
Joining Date	Date

Step 6: Student Validation Rules

1. Roll Number Validation

Rule Name

Roll_Number_Validation

Formula

Roll_Number__c <= 0

Error Message

Roll Number must be greater than 0

2. Email Validation

Rule Name

Email_Validation

Formula

```
NOT(CONTAINS(Email__c, "@"))
```

Error Message

Email must contain @ symbol

3. Student Name Validation

Rule Name

Student_Name_Validation

Formula

```
ISBLANK(Name)
```

Error Message

Student name cannot be blank

4. Marks Validation

Rule Name

Marks_Validation

Formula

```
OR(  
  Marks__c < 0,  
  Marks__c > 100  
)
```

Error Message

Marks should be between 0 and 100

5. Joining Date Validation

Rule Name

Joining_Date_Validation

Formula

```
Joining_Date__c > TODAY()
```

Error Message

Joining Date cannot be future date

Step 7: Faculty Validation Rules

1. Faculty Name Validation

Rule Name

Faculty_Name_Validation

Formula

```
LEN(Name) < 3
```

Error Message

Faculty Name must contain at least 3 characters

2. Faculty ID Validation

Rule Name

Faculty_ID_Validation

Formula

```
Faculty_ID__c <= 0
```

Error Message

Faculty ID must be greater than 0

3. Make Faculty ID Unique

Go to: Faculty → Fields & Relationships → Faculty ID → Edit

Check:

- Unique

Save.

4. Salary Validation

Rule Name

Salary_Validation

Formula

```
OR(  
Salary__c <= 10000,  
Salary__c >= 500000  
)
```

Error Message

Salary must be between 10,000 and 500,000

5. Faculty Joining Date Validation

Rule Name

Faculty_Joining_Date_Validation

Formula

```
Joining_Date__c > TODAY()
```

Error Message

Joining Date cannot be future date

Step 8: Create Tabs

Problem Faced

Student and Faculty objects were not appearing in Navigation Items.

Reason

Tabs were not created.

Solution

Go to: Setup → Tabs → New

Create:

- Student Tab
- Faculty Tab

Choose any tab style.

Save.

Step 9: Create Lightning App

Go to: Setup → App Manager → New Lightning App

App Name

College Management

Continue clicking:

- Next
 - Next
 - Next
-

Step 10: Navigation Items Problem

Error Faced

Students and Faculties were not visible in Navigation Items.

Solution

After creating tabs: 1. Refresh page 2. Search Students 3. Add using Arrow button 4. Add Faculties 5. Click Next 6. Save & Finish

Step 11: App Not Opening Problem

Error Faced

College Management app not opening.

Reason

Search was performed inside Setup instead of App Launcher.

Correct Method

1. Click 9 Dots (App Launcher)
 2. Search College Management
 3. Open app
-

Step 12: Final Working Output

Successfully Working:

- Student Object
 - Faculty Object
 - Validation Rules
 - Lightning App
 - Tabs
 - Record Forms
-

Step 13: Validation Testing

Student Validation Tests

Test	Wrong Value	Expected Result
Roll Number	0	Error
Email	abcm@gmail.com	Error
Marks	120	Error
Joining Date	Future Date	Error

Faculty Validation Tests

Test	Wrong Value	Expected Result
Faculty Name	ab	Error
Faculty ID	0	Error
Duplicate Faculty ID	Existing ID	Error
Salary	5000	Error
Joining Date	Future Date	Error

Final Conclusion

The College Management Lightning Application was successfully implemented with:

- Student Management
- Faculty Management
- Validation Rules
- Unique Faculty ID Validation
- Lightning App Navigation
- Salesforce Lightning Experience
- Data Accuracy Validation System

Project Status: COMPLETED SUCCESSFULLY